

Prerequisites for Canonical Form.

Def. Let $T: V \rightarrow V$ be a linear operator on V .

A subspace $W \subseteq V$ is called T -invariant if: $T[W] \subseteq W$.

Note: $\text{Im } T$, $\ker T$, E_λ are all T -invariant.

Lemma. A T -cyclic subspace generated by $x \in V$

$$W = \langle T^n(x) \mid n = 0, 1, 2, \dots \rangle$$

is the smallest T -invariant containing $x \in V$.

Proof. T -invariant is clear. Suppose that W' is a T -invariant subspace containing $x \in V$.

Then, $x \in W' \Rightarrow T(x) \in W' \Rightarrow T^2(x) \in W' \dots$

By induction, for all $n \in \mathbb{N}$, $T^n(x) \in W'$. That is, W' contains generating set of W .

Lemma. If W is T -invariant, then also $(T - cI)$ -invariant, for any $c \in F$.

Proof. Let $x \in W$ be given. Then,

$$(T - cI)(x) = \underbrace{T(x)}_{\in W} - c \underbrace{x}_{\in W} \in W$$

Canonical Form.

Def. Let $T: V \rightarrow V$ be a linear operator, and $\lambda \in F$ be a scalar.

A non-zero vector $x \in V$ is called Generalized eigenvector of T corresponding to λ if;

$$\text{for some } p \in \mathbb{N}, (T - \lambda I)^p(x) = 0$$

Denote a set of generalized eigenvectors of λ as K_λ . ($K_\lambda = \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n$)

Note: If a scalar $\lambda \in F$ has a generalized eigenvector s.t. $(T - \lambda I)^p(0) = 0$, then λ is eigenvector of T , corresponding the eigenvector $(T - \lambda I)^{p-1}(x)$.

Thm. Let $T: V \rightarrow V$ be a linear operator, and $\lambda \in F$ be an eigenvalue of T .

1) Generalized Eigenspace K_λ is T -invariant subspace which contains E_λ .

2) For any eigenvalue $\mu \in F$ with $\mu \neq \lambda$, $K_\mu \cap K_\lambda = \{0\}$.

3) For any scalar $\gamma \in F$ with $\gamma \neq \lambda$, the restriction

$$(T - \lambda I)|_{K_\lambda}: K_\lambda \rightarrow K_\lambda; x \mapsto (T - \lambda I)(x)$$

is bijection (That is, Isomorphism)

Proof Clearly, $0 \in K_\lambda$. To use subspace criterion, let $c, \gamma \in K_\lambda$ be given.

That is, for some $p, q \in \mathbb{N}$, $(T - \lambda I)^p(x) = (T - \lambda I)^q(y) = 0$. Now,

$$\begin{aligned} (T - \lambda I)^{p+q}(cx + \gamma) &= c \cdot (T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(\gamma) \\ &= c \cdot (T - \lambda I)^q(0) + (T - \lambda I)^p(0) \end{aligned}$$

This implies $cx + \gamma \in K_\lambda$.

To show that T -invariant, let $x \in K_\lambda$ s.t. $(T - \lambda I)^p(x) = 0$. Then

$$(T - \lambda I)^p(T(x)) = T \circ (T - \lambda I)^p(x) = T(0) = 0.$$

This implies $T[K_\lambda] \subseteq K_\lambda$

Let $\mu \in F$ be an eigenvalue of T s.t. $\mu \neq \lambda$. Choose $x \in K_\mu \cap K_\lambda$. Then, for some $p, q \in \mathbb{N}$,

$$(T - \lambda I)^p(x) = (T - \mu I)^q(x) = 0.$$

In other words, for $f_\lambda(t) = (t - \lambda)^p$, $f_\mu(t) = (t - \mu)^q$, $f_\lambda(T)(x) = f_\mu(T)(x) = 0$.

Since $F[t]$ is PID, it is also Bezout domain, there exist $g_1(t), g_2(t) \in F[t]$ s.t.

$$g_1(t)f_\lambda(t) + g_2(t)f_\mu(t) = \gcd(f_\lambda(t), f_\mu(t)) = 1.$$

Therefore, $g_1(T)f_\lambda(T)(x) + g_2(T)f_\mu(T)(x) = g_1(T)(0) + g_2(T)(0) = 0 = x$.

Let $\gamma \in F$ with $\gamma \neq 1$ be given. Since K_λ is T -invariant, for any $x \in K_\lambda$,

$$(T - \gamma I)(x) = T(x) - \gamma \cdot x \in K_\lambda$$

That is, K_λ is $(T - \gamma I)$ -invariant.

Let $x \in K_\lambda$, and suppose $(T - \gamma I)(x) = 0$. Then, $T(x) = \gamma x$, thus $x \in E_\gamma \subseteq K_\gamma$

But by 2., $K_\gamma \cap K_\lambda = \{0\}$, $x = 0$. That is, $\ker(T - \gamma I)|_{K_\lambda} = \{0\}$, thus $1-1$.

Let $x \in K_\lambda$ be given. put $p \in \mathbb{N}$ be the smallest Natural number s.t. $(T - \lambda I)^p(x) = 0$.

Define $W = \langle \{x, (T - \lambda I)(x), \dots, (T - \lambda I)^{p-1}(x)\} \rangle$

Then, W equals to $(T - \lambda I)$ -cyclic subspace generated by x ,

thus W is the smallest $(T - \lambda I)$ -invariant subspace containing x , thus $W \subseteq L_\lambda$.

Also, W is $(T - \gamma I)$ -invariant. Now, a restriction

$$(T - \gamma I)|_W : W \rightarrow W : \gamma \mapsto (T - \gamma I)(\gamma)$$

is well-defined injection, and surjection, since W is fin. dim.

Finally, there exists $\gamma \in W$ s.t. $(T - \gamma I)|_W(\gamma) = x$, thus $(T - \gamma I)|_{K_\lambda}$ is surjection

Thm. Let $T: V \rightarrow V$ be a linear operator on fin. dim V over F .

If a characteristic polynomial $\det(T - tI)$ splits over F ,

then for an eigenvalue λ with multiplicity $m \geq 1$, the followings are hold:

1) $\dim K_\lambda \leq m$

2) $K_\lambda = \ker(T - \lambda I)^m$

Proof. Denote $f(t) = \det(T - tI)$ and $h(t) = \det(T_{K_\lambda} - tI)$.

If T_{K_λ} has another eigenvalue μ , then there exists a non-zero vector v s.t.

$$v \in E_\mu \subseteq K_\mu \text{ and } v \in K_\lambda.$$

But, since $K_\mu \cap K_\lambda = \{0\}$ it is contradiction. Now,

$$h(t) = (-1)^d (t - \lambda)^d$$

where $d = \dim K_\lambda$. Since $h(t)$ divides $f(t)$, $\dim K_\lambda = d \leq m$.

Since $K_\lambda = \bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n$, $\ker(T - \lambda I)^m \subseteq K_\lambda$ is trivial.

And, $f(t) = (t - \lambda)^m g(t)$ where $g(t)$ has no factor $(t - \lambda)$.

For any $\gamma \neq \lambda$, $(T - \gamma I)|_{K_\lambda}$ is bijection. Thus, $g(T)|_{K_\lambda}$ is bijection.

Let $x \in K_\lambda$. Then, for some $y \in K_\lambda$, $x = g(T)(y)$. Now,

$$(T - \lambda I)^m(x) = (T - \lambda I)^m(g(T)(y)) = f(T)(y) = 0.$$

by Cayley-Hamilton Theorem. Thus $x \in \ker(T - \lambda I)^m$.

Thm. Let V be a fin. dim vector space, $T: V \rightarrow V$ be a linear operator.

If a characteristic polynomial $f(t)$ of T splits over F ,
then for distinct eigenvalues $\lambda_1, \dots, \lambda_k$ of T ,

$$V = K_{\lambda_1} \oplus \dots \oplus K_{\lambda_k}.$$

Proof. Denote $n = \dim V$, and

$$f(t) = (-1)^n (t - \lambda_1)^{n_1} \dots (t - \lambda_k)^{n_k}.$$

And for each $1 \leq j \leq k$, define

$$f_j(t) = \prod_{\substack{i \neq j \\ 1 \leq i \leq k}} (t - \lambda_i)^{n_i}$$

Since f_j ($1 \leq j \leq k$) are mutually disjoint, there exist $g_j(t)$ ($1 \leq j \leq k$) such that

$$g_1(t)f_1(t) + \dots + g_k(t)f_k(t) = 1.$$

Now, for any $x \in V$,

$$g_1(T)f_1(T)(x) + \dots + g_k(T)f_k(T)(x) = x.$$

Meanwhile for each $g_j(T)f_j(T)(x)$, the Cayley-Hamilton thm gives

$$\begin{aligned} & (T - \lambda_j I)^{n_j} (g_j(T)f_j(T)(x)) \\ &= (T - \lambda_j I)^{n_j} (f_j(T)g_j(T)(x)) \\ &= f(T)(g_j(T)(x)) \\ &= 0 \end{aligned}$$

That is, $g_j(T)f_j(T)(x) \in K_{\lambda_j}$. This proves $V = K_{\lambda_1} + \dots + K_{\lambda_k}$.

To show the uniqueness, note that since K_{λ_i} is T -invariant, $f_j(T)$ -invariant.

Also, a restriction $f_j(T)|_{K_{\lambda_j}}$ is bijection, being composition of bijections.

$$\text{Moreover, for } i \neq j, \quad f_j(T)[K_{\lambda_i}] = f_j(T)[\ker(T - \lambda_i I)^{n_i}] = \{0\}$$

Now, if $\sum_{i=1}^k v_i = \sum_{i=1}^k w_i$ where $v_i, w_i \in K_{\lambda_i}$ ($1 \leq i \leq k$), then for each $1 \leq j \leq k$,

$$f_j(T)(v_j) = \sum_{i=1}^k f_j(T)(v_i) = \sum_{i=1}^k f_j(T)(w_i) = f_j(T)(w_j).$$

Since $f_j|_{K_{\lambda_j}}$ is injection, $v_j = w_j$.

Thm. Let V be a finite dimensional vector space, $T: V \rightarrow V$ be a linear operator.
If $\det(T-tI)$ splits over F as

$$\det(T-tI) = (\lambda_1 - t)^{m_1} (\lambda_2 - t)^{m_2} \cdots (\lambda_k - t)^{m_k}.$$

Then, $V = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_k}$. Let β_{λ_i} be a basis for K_{λ_i} .

The following statements are true:

1) $i \neq j \Rightarrow \beta_i \cap \beta_j = \emptyset$.

2) $\beta = \bigcup_{i=1}^k \beta_i$ is basis for V

3) for each $1 \leq i \leq k$, $\dim K_{\lambda_i} = m_i$.

Proof. 1) and 2) are clear. To show 3), see that:

$$\dim V = \sum_{i=1}^k m_i = \sum_{i=1}^k \dim(K_{\lambda_i})$$

And for all $1 \leq i \leq k$, $\dim K_{\lambda_i} \leq m_i$, thus $m_i = \dim K_{\lambda_i}$.

Def. Let $T: V \rightarrow V$ be a linear operator, and $x \in V$ be a generalized eigenvector of T corresponding to $\lambda \in F$. For the smallest $p \in \mathbb{N}$ s.t. $(T - \lambda I)^p(x) = 0$, a set

$$\{(T - \lambda I)^{p-1}(x), (T - \lambda I)^{p-2}(x), \dots, (T - \lambda I)(x), x\}$$

is called cycle of x , and denote C_x . (Note: only $(T - \lambda I)^{p-1}(x)$ is eigenvector of $T - \lambda I$).

Thm. Let $T: V \rightarrow V$ be a linear operator, $x \in V$ be a general. eigen. of T corresponding to $\lambda \in F$.

For cycle C_x of x , $\langle C_x \rangle$ is T -invariant, and $[T|_{\langle C_x \rangle}]_{C_x}$ is Jordan block.

Proof. Denote $C_x = \{(T - \lambda I)^{p-1}(x), \dots, x\}$.

For each $0 \leq i \leq p-1$,

$$(T - \lambda I)((T - \lambda I)^i(x)) = (T - \lambda I)^{i+1}(x) \in \langle C_x \rangle$$

Thus, $\langle C_x \rangle$ is $(T - \lambda I)$ -invariant, so that T -invariant.

Meanwhile, $[T|_{\langle C_x \rangle}]_{C_x} = [T - \lambda I|_{\langle C_x \rangle}]_{C_x} + [\lambda I|_{\langle C_x \rangle}]_{C_x}$ and

$$[T - \lambda I|_{\langle C_x \rangle}]_{C_x} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 \end{bmatrix}$$

Gives the result.

Note: $C_x = \{(T - \lambda I)^{p-1}(x), \dots, x\}$ are distinct, because

'If $0 \leq i < j < p-1$ satisfies

$$(T - \lambda I)^i = (T - \lambda I)^j$$

$$\Rightarrow (T - \lambda I)^{\underbrace{p-j-1+i}_{p-1-(j-i)}} = (T - \lambda I)^{p-1} = 0$$

Contradiction.

Thm. Let $\lambda \in F$ be an eigenvalue of $T: V \rightarrow V$.

Let $x_1, \dots, x_b \in K_\lambda$ be distinct vectors.

If initial vector of each C_{x_i} are mutually different and initial vectors are linearly independent

then C_{x_i} are mutually disjoint and $C = \bigcup_{i=1}^b C_{x_i}$ is linearly independent.

Proof. Denote that

$$C_x = \{(T - \lambda I)^{a-1} x, \dots, x\}, \quad C_y = \{(T - \lambda I)^{b-1} y, \dots, y\} \quad a \leq b$$

If for some $1 \leq i \leq a$ $1 \leq j \leq b$

$$(T - \lambda I)^{a-i} x = (T - \lambda I)^{b-j} y,$$

$$\text{Then } (T - \lambda I)^a x = (T - \lambda I)^{b-j+i} y = 0 = (T - \lambda I)^{a-i+j} x$$

If $i-j=0$, then these have same initial vector, contradiction.

If $i-j > 0$, then contradiction to minimality.

$$\begin{pmatrix} a & & \\ & b & \\ & & c \\ & & & \ddots \end{pmatrix}$$

$$\begin{pmatrix} a & 1 & & \\ & b & 1 & \\ & & c & 1 \\ & & & \ddots \end{pmatrix}$$

Let $T: V \rightarrow V$ be a linear operator, and $f(T)$ be a characteristic polynomial of T .

Def. Suppose there $\lambda \in F$ be an eigenvalue of T . Define a set:

$$K_\lambda := \left\{ x \in V \mid (T - \lambda I)^p(x) = 0 \text{ for some } p \in \mathbb{N} \right\} = \bigcup_{n=1}^{\infty} \text{Ker}(T - \lambda I)^n$$

It is called a generalized eigenspace corresponding to λ .

For non-zero $x \in K_\lambda$, the smallest positive integer $p \in \mathbb{Z}$ s.t. $(T - \lambda I)^p(x) = 0$ is called period of x .
(If $x=0$, period of x is assigned as zero.) (or length)

Lemma. Let $\lambda \in F$ be an eigenvalue of T . Then, the following properties are satisfied:

- 1). K_λ is T -invariant subspace of V containing E_λ .
- 2). If $\mu \in F$ is an eigenvalue of T s.t. $\mu \neq \lambda$, then $K_\mu \cap K_\lambda = \{0\}$
- 3). For any CGF s.t. $c \in F$, $(T - cI)|_{K_\lambda}$ is a bijection.

Proof. Let $x, y \in K_\lambda$ and $\lambda \in F$ be given. Denote $p, q \in \mathbb{Z}$ as period of x, y , respectively.

Then,

$$(T - \lambda I)^{p+q}(\alpha x + y) = \alpha (T - \lambda I)^{p+q}(x) + (T - \lambda I)^{p+q}(y) = 0.$$

Clearly, $0 \in K_\lambda$. Therefore, subspace criterion gives $K_\lambda \leq V$.

To show T -invariant, let $x \in K_\lambda$. Then, for period $p \in \mathbb{Z}$ of x ,

$$(T - \lambda I)^p(T(x)) = ((T - \lambda I)^p \circ T)(x) = (T \circ (T - \lambda I)^p)(x) = T(0) = 0.$$

Thus $T(x) \in K_\lambda$. Including E_λ is clear.

Show 2). Let $w \in K_\mu \cap K_\lambda$ be given. Let $p, q \in \mathbb{Z}$ be period of w for μ and λ , respectively.

Since $\mu \neq \lambda$, $(t - \mu)^p, (t - \lambda)^q \in F[t]$ are disjoint. Now, there exist $g_1(t), g_2(t) \in F[t]$ s.t.

$$g_1(t)(t - \mu)^p + g_2(t)(t - \lambda)^q = 1.$$

because $F[t]$ is P.I.D. Now,

$$(g_1(T)(T - \mu)^p + g_2(T)(T - \lambda)^q)(w) = 0 = w$$

Thus $K_\mu \cap K_\lambda = \{0\}$.

Finally, show 3). Since for any $w \in K_\lambda$, $(T - cI)(w) = 0 \Rightarrow w \in E_c \subseteq K_c$

By 2), $w = 0$. That is, kernel of $(T - cI)|_{K_\lambda}$ is zero space, thus injective.

To show surjective, let $x \in K_\lambda$. For period $p \in \mathbb{Z}$ of x ,

$$W = \{x, (T - \lambda I)(x), \dots, (T - \lambda I)^{p-1}(x)\} \subseteq K_\lambda$$

is $(T - \lambda I)$ -invariant. Now, $(T - \lambda I)|_W$ is injection on fin. dim space, thus surjection.

Now, we can find $y \in W$ s.t. $(T - \lambda I)(y) = x$.

Thm. If $f(t)$ splits over F , then for any eigenvalue λ with multiplicity $m \in \mathbb{N}$,

$$\dim K_\lambda \leq m \text{ and } K_\lambda = \ker((T - \lambda I)^m)$$

proof. Let $h(t)$ be a characteristic polynomial of a restriction $T|_{K_\lambda}$.

Since for any distinct eigenvalue $\mu \neq \lambda$, $K_\mu \cap K_\lambda = \{0\}$,

λ is a unique eigenvalue of $T|_{K_\lambda}$. Moreover, since $h(t)$ divides $f(t)$,

$$h(t) = (t - \lambda)^d \quad (d \leq m).$$

Clearly, $\ker(T - \lambda I)^m \subseteq K_\lambda$.

Meanwhile, by assumption, $f(t) = (t - \lambda)^m \cdot g(t)$ where $g(\lambda) \neq 0$.

Therefore, $g(T)|_{K_\lambda}$ is bijective. Now, for any $x \in K_\lambda$, there is $y \in K_\lambda$ s.t. $g(T)(y) = x$.

The Cayley-Hamilton Thm gives

$$(T - \lambda I)^m(x) = (T - \lambda I)^m(g(T)(y)) = f(T)(y) = T_0(y) = 0.$$

Thus $x \in \ker(T - \lambda I)^m$.

$$\text{Im } U = (T - \lambda I)|_{K_\lambda} [K_\lambda]$$

$$= (T - \lambda I)|_{K_\lambda} \left[\bigcup_{n=1}^{\infty} \ker(T - \lambda I)^n \right]$$

$$= \bigcup_{n=1}^{\infty} (T - \lambda I) [\ker(T - \lambda I)^n]$$

$$\begin{array}{c} V \\ | \\ K_\lambda \\ | \\ \text{Im } U \end{array}$$

$$T|_{\text{Im } U}$$